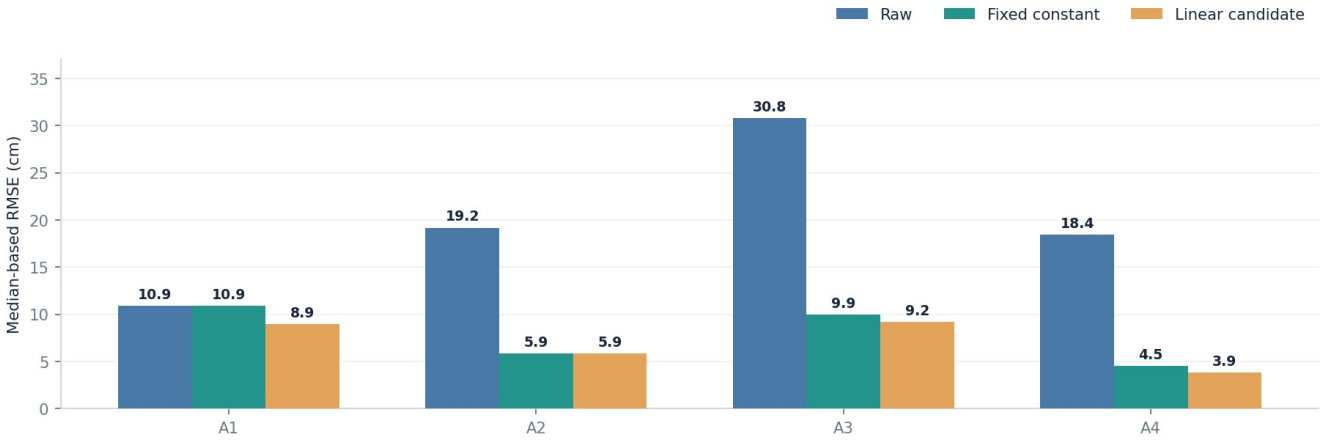
TECHNICAL REPORT / CALIBRATION RESULTS

UWB Ranging Calibration

Fixed Per-Anchor Constant Offset vs. Linear Correction | Delta RTLS

RAW RMSE	FIXED CONSTANT	LINEAR CANDIDATE
21.14 cm	**8.36 cm**	**7.40 cm**
Before calibration	60.4% RMSE reduction	In-sample fit only

Before vs. After: RMSE by Anchor



Anchor	Final fixed offset	Raw RMSE	Fixed constant RMSE	Linear fit RMSE
A1	+0.00 cm	10.90 cm	10.90 cm	8.94 cm
A2	+18.28 cm	19.17 cm	5.88 cm	5.87 cm
A3	+28.97 cm	30.76 cm	9.94 cm	9.18 cm
A4	+17.87 cm	18.44 cm	4.53 cm	3.86 cm

Calibration Decision

Retain the current per-anchor constant profile as the primary engineering result: A1 = 0.0000 m, A2 = +0.1828 m, A3 = +0.2897 m, A4 = +0.1787 m. Linear correction provides a lower in-sample residual, but its cross-position performance does not justify replacing the fixed profile.

Ranging RMSE values above use per-position median distances. Positioning performance below is assessed separately from synchronized range cycles.

Positioning Assessment Against Proposal Criteria

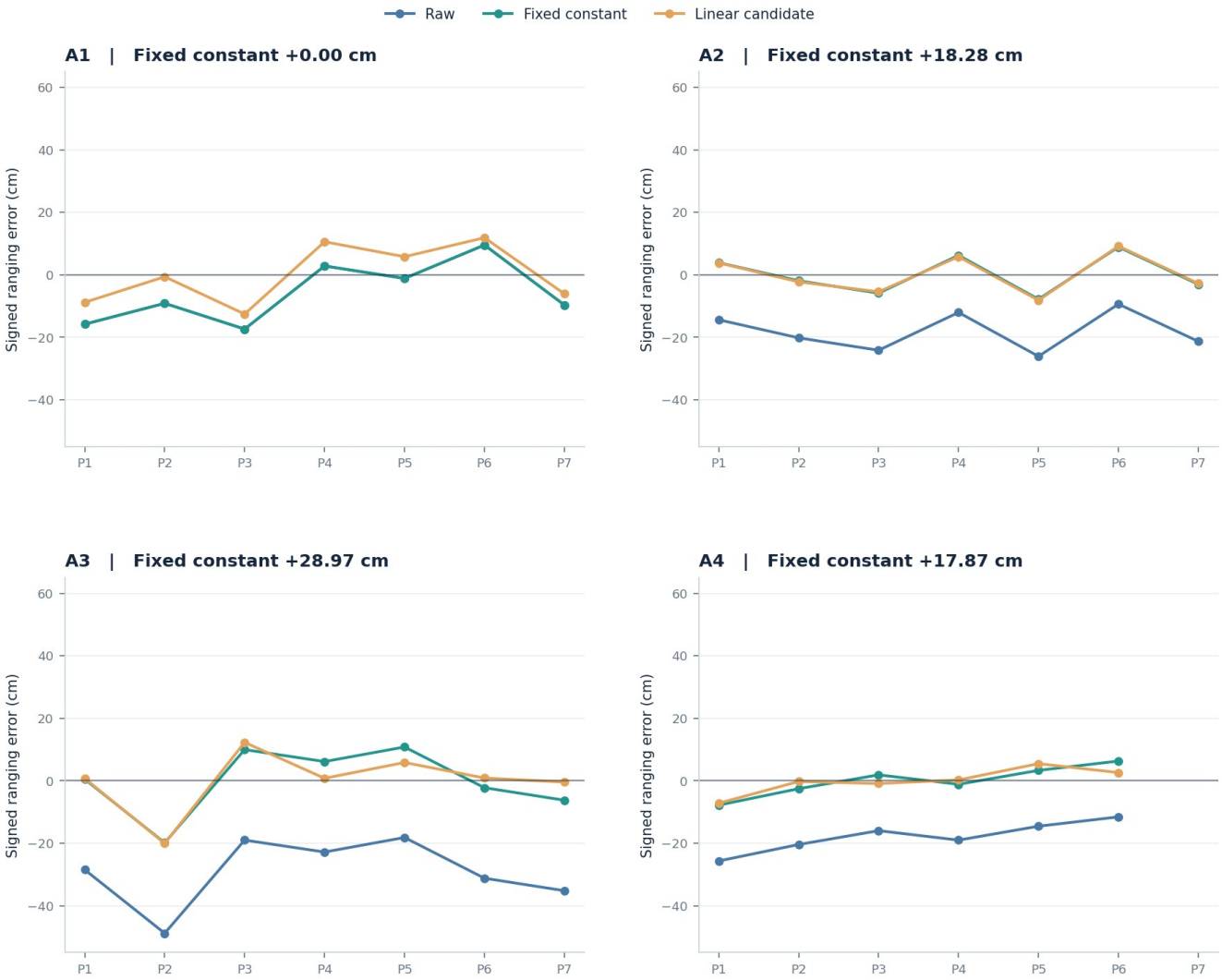
Assessment	Proposal target	RAW	Constant	Linear	Result
Median position error	≤ 30 cm	14.94 cm	8.36 cm	7.84 cm	**PASS**
P95 position error	≤ 50 cm	26.67 cm	22.65 cm	22.98 cm	**PASS**

Assessment result: both thresholds are met numerically in offline 2D multilateration (2,441 synchronized cycles); independent validation and live positioning remain pending.

ERROR DISTRIBUTION / BEFORE VS AFTER

Signed Ranging Error Across Positions

Curves show the difference between measured and reference distance. Values closer to zero indicate less ranging bias; negative values indicate underestimation.



P1 (0.75, 0.25) P2 (0.50, 0.25) P3 (1.00, 0.25) P4 (0.50, 0.10)

P5 (0.50, 0.40) P6 (1.00, 0.10) P7 (1.00, 0.40) | Coordinates in meters

Observations

A1 retains its raw baseline because its error changes sign between positions. A2 benefits strongly from constant correction. A3 improves, but residuals remain position-dependent. The fixed A4 correction substantially reduces the error within the retained measurements.

MODEL SELECTION / GENERALIZATION

Constant vs. Linear: Cross-Position Check

Leave-one-position-out evaluation re-fits each candidate on the remaining positions and tests the held-out position. Lower RMSE indicates better transfer to unseen coordinates within this dataset.

RE-FITTED CONSTANT

9.13 cm

Leave-one-position-out RMSE

RE-FITTED LINEAR

10.30 cm

Leave-one-position-out RMSE



Anchor	Constant CV RMSE	Linear CV RMSE	Preferred by CV
A1	10.72 cm	13.71 cm	Constant
A2	6.85 cm	8.39 cm	Constant
A3	11.60 cm	11.20 cm	Linear (slight)
A4	5.43 cm	5.45 cm	Comparable

Exploratory Linear Candidate: d_linear = a × d_raw + b

Anchor	Slope a	Intercept b (m)
A1	0.898826	+0.137824
A2	0.983743	+0.193213
A3	0.831315	+0.384110
A4	0.896984	+0.243786

The cross-position constant values above are re-fitted in each fold for a fair model-class comparison; they are not the frozen operational offsets. The cross-position check reuses the calibration data and is not an independent field validation.

IMPLEMENTATION GUIDANCE / CALCULATION

Monitoring Considerations

Signal strength: RX power provides a useful quality indicator, but strong received power does not guarantee accurate ranging. Interpret power together with measurement consistency and other available receiver diagnostics.

Variable bias: Residual ranging bias may change with position, orientation and local radio propagation conditions. A3 is especially sensitive to this variation in the current results.

Data quality: Monitor age, temporal variation, disagreement among anchors and solver residuals before accepting a measurement for multilateration.

Profile integrity: Store raw and corrected distances separately, together with the applied profile identifier and coefficient version, to prevent double correction.

Validation: Freeze the selected coefficients and evaluate them on new stationary locations and real-time X/Y solver outputs before claiming operational positioning accuracy.

Calculation Method

1. Three-dimensional reference distance

$$d_{true} = \sqrt{(x_{tag} - x_{anchor})^2 + (y_{tag} - y_{anchor})^2 + (z_{tag} - z_{anchor})^2}$$

Reference distance includes the 0.25 m anchor-to-tag height difference.

2. Fixed per-anchor constant offset

$$d_{constant,i,j} = d_{raw,i,j} + c_i$$

c_i is the frozen final offset shown in the Results table. A1 intentionally remains uncorrected ($c_{A1} = 0$).

3. Linear candidate (exploratory)

$$d_{linear,i,j} = a_i \times d_{raw,i,j} + b_i$$

Slope a_i and intercept b_i were fitted using least squares from per-position raw medians and known reference distances.

4. Root mean squared ranging error

$$RMSE = \sqrt{\frac{1}{N} \times \Sigma(d_{estimated,i,j} - d_{true,i,j})^2}$$

The same retained observations are used for each method within each anchor. Summary values are computed on a common set of per-position medians, with equal weight per retained observation.

5. Percentage improvement

$$\text{RMSE reduction (\%)} = (RMSE_{raw} - RMSE_{constant}) / RMSE_{raw} \times 100$$

$$= (21.14 - 8.36) / 21.14 \times 100 \approx 60.4\%$$

Status: candidate calibration supported by offline ranging analysis. Independent verification and real-time positioning tests remain outstanding.